Hybrid versus standard tensor trains under time evolution: benchmark of the explicit solver with SVD truncation

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The hybrid tensor-train encoding is benchmarked against a standard quantized tensor train for the time evolution of a passive scalar, at four resolutions from 256^2 to 2048^2. Both encodings are advanced by the same explicit solver: exact MPO–MPS contraction followed by a two-sweep SVD truncation. At a fixed target accuracy the hybrid encoding reaches that accuracy with a factor 1.8–2.5 fewer parameters, and the factor is constant across the resolutions tested. The head size that achieves it holds half the bits of each axis, $N \simeq L$, corresponding to a $2^{L/2} \times 2^{L/2}$ coarse grid represented densely and the finer scales carried by the tail. The error made in evolving tracks the error of merely representing the same field within a factor 1.5–25 for every encoding, resolution and budget measured. Two further flows are reported: a forced cellular flow, in which the rank plateau found in the decaying cascade persists in a statistically steady state, and a frozen DNS velocity snapshot, under which the rank also plateaus but the evolution error ceases to be monotone in the bond cap while the representation error remains so.

I. CONVENTIONS: WHAT χ, ϵ AND ρ REFER TO

Three symbols recur throughout, and each of them can be applied to the state θ or to any of the coefficient fields $\mathbf{u}$, ψ and f. Only one of them is a control variable. Table I fixes the correspondence, and every later use of a bare χ, ϵ or ρ refers to θ.

	applied to θ (the state)	applied to $\mathbf{u}, \psi, f$ (the coefficients)
χ largest bond dimension	**the swept control** (`bd_first` = `bd_remaining`); binding in every production run, i.e. measured $\chi = \chi_{\text{cap}}$	fixed per object, never swept: 4 (cellular $\mathbf{u}$), 34 (DNS $\mathbf{u}$), 2 (f)
ϵ relative SVD discard threshold	**off** (ϵ =`None`) in all 101 production runs; recorded as -1 in the run files	fixed: 10^{-13} (ψ cellular, f), 10^{-2} (ψ DNS), 10^{-10} after the derivative MPO
ρ stored parameters divided by 4^L	**an output**, determined by (N, χ) — not an input	output; $\rho(\mathbf{u}) = 1.1 \times 10^{-3}$ to 1.9×10^{-2} (cellular), 0.14 (DNS)

TABLE I. The three symbols by object. The bond cap χ on θ is the only quantity swept; the SVD threshold ϵ on θ is disabled in the production runs, so truncation is governed by the cap alone; and ρ is measured after the fact.

Two consequences are worth stating explicitly. First, because ϵ is off and the cap is reached at every bond, the measured χ equals the requested cap in all 101 production runs, and ρ is a deterministic function of (N, χ) rather than something the solver chooses. Second, ϵ still appears in four fixed roles that are *not* the state truncation: encoding ψ, truncating $\mathbf{u}$ after the derivative MPO is applied to ψ, encoding f, and the diagnostic probes described below. Quoting an ϵ without naming its object leaves these indistinguishable.

The compression ratio is $\rho = P/4^L$ with P the number of stored parameters, so $\rho = 1$ is the cost of the dense field and $\rho > 1$ is possible — the DNS streamfunction reaches $\rho = 1.33$ at $L = 9$. For the hybrid encoding P counts the dense head together with the tail cores.

The *intrinsic rank* reported in several places is a property of the exact solution, obtained by recompressing the dense field at a probe accuracy with no feedback into the dynamics. It is always quoted with that probe ϵ and with the head size N it was measured in, since both change it: in the forced steady state the same field gives 58.9, 41.7 and 23.0 at $N = 0$, 8 and 10 when probed at $\epsilon = 10^{-6}$, and 41.7, 20.0 and 6.0 at $N = 8$ when probed at 10^{-6}, 10^{-4} and 10^{-2}. A rank is also reported against the geometric bound $\min(2^{\text{cut}}, 2^{2L-\text{cut}})$ at the same cut, without which a saturated measurement reads as a property of the field rather than of the grid.

The Batchelor scale is $\ell_B = \sqrt{\kappa/\gamma}$, with γ the rms strain rate $(2\langle S_{ij}S_{ij}\rangle)^{1/2}$ — equal, for an incompressible two-dimensional field, to $\langle|\nabla\mathbf{u}|^2\rangle^{1/2}$ and to the rms vorticity, all three measured and agreeing to four digits. It is the smallest scale the scalar populates, so $\ell_B/\Delta x$ states how well that scale is resolved: holding it fixed while L grows makes a finer grid add scales rather than over-resolve the ones already present.

Table II collects every knob of every run family.

flow	grid	κ	$\ell_B/\Delta x$	T	steps	N	$\chi(\theta)$	$\rho(\theta)$	$\chi(\mathbf{u})$
cellular, decaying	256^2	5×10^{-3}	2.88	8	1630	$0,4,6,8,10$	$4,6,8,12,16,24,32$	0.0208–0.2708	4
cellular, decaying	256^2	5×10^{-3}	2.88	16	3259	6	$16,32,64,128$	0.0677–0.6458†	4
cellular, decaying	512^2	1.25×10^{-3}	2.88	8	3259	$0,8,10$	$12,16,24,32$	0.0131–0.1536	4
cellular, decaying	1024^2	3.125×10^{-4}	2.88	8	6519	$0,8,10,12$	$16,24,32,48,64$	0.0065–0.2708	4
cellular, decaying	2048^2	7.8125×10^{-5}	2.88	8	13038	$0,10,12$	$24,32,48,64$	0.0038–0.0716	4
frozen DNS	512^2	1.5×10^{-2}	2.61	2	1992	$0,8,10,12$	$16,24,32,48$	0.0221–0.7669	34
cellular, forced ($f=0.8\sin 4y$)	256^2	5×10^{-3}	2.88	60	12222	$0,6,8,10$	$16,24,32$	0.0677–0.5521	4

TABLE II. All 101 production runs, by family. $\chi(\theta)$ is the swept control and $\rho(\theta)$ the range it produces across the head sizes listed; $\epsilon(\theta)$ is off throughout except in one run of the family marked †, where a cap of 128 was left loose and $\epsilon=10^{-6}$ was allowed to select the bond instead, giving $\chi=61$; the other three runs of that family use caps 16, 32 and 64 with ϵ off. Δt is CFL-limited in every family. $\gamma=1$ for the cellular flow and 14.6 for the truncated DNS velocity, which is why the two reach a comparable $\ell_B/\Delta x$ at very different κ.

II. SETUP

The benchmark problem is a decaying passive scalar on a periodic square,

$$\partial_t\theta+\mathbf{u}\cdot\nabla\theta=\kappa\nabla^2\theta,\qquad \nabla\cdot\mathbf{u}=0, \tag{1}$$

with no forcing. Multiplying Eq. (1) by θ gives a pointwise statement about $\theta^2/2$,

$$\partial_t\frac{\theta^2}{2}+\mathbf{u}\cdot\nabla\frac{\theta^2}{2}=\kappa\,\theta\nabla^2\theta, \tag{2}$$

and two rewritings turn it into a local conservation law. Incompressibility makes the transport term a pure divergence, since $\mathbf{u}\cdot\nabla(\theta^2/2)$ and $\nabla\cdot(\mathbf{u}\,\theta^2/2)$ differ by $(\theta^2/2)\,\nabla\cdot\mathbf{u}=0$; and the product rule splits the diffusive term as $\kappa\theta\nabla^2\theta=\kappa\nabla\cdot(\theta\nabla\theta)-\kappa|\nabla\theta|^2$. Hence

$$\partial_t\frac{\theta^2}{2}+\nabla\cdot\Big(\mathbf{u}\frac{\theta^2}{2}-\kappa\theta\nabla\theta\Big)=-\kappa|\nabla\theta|^2, \tag{3}$$

and averaging over the periodic box annihilates the divergence, leaving

$$\frac{d}{dt}\frac{\langle\theta^2\rangle}{2}=-\kappa\langle|\nabla\theta|^2\rangle. \tag{4}$$

Advection therefore enters only inside a divergence: it redistributes variance in space and across scales without creating or destroying any, and the whole loss is the sign-definite diffusive term, whose size is set by the gradient the cascade has built. With a source the right-hand side gains $\langle\theta f\rangle$, and the forced runs equilibrate where the two balance. Eq. (4) holds discretely only to the extent that the discrete transport operator keeps the skew-symmetry that made the term a divergence, which is why any variance unaccounted for by $\kappa\langle|\nabla\theta|^2\rangle$ in Fig. 1 is numerical. The velocity is the steady cellular flow $\mathbf{u}=(\sin x\cos y,\,-\cos x\sin y)$, obtained from the streamfunction $\psi=\sin x\,\sin y$. It is constructed by applying the solver's own derivative MPOs to the encoded ψ, which makes it divergence-free to 6×10^{-13} at the discrete level (9×10^{-11} at $L=11$) and gives it $\chi(\mathbf{u})=4$ at every resolution and head size, at a cost $\rho(\mathbf{u})$ from 1.9×10^{-2} at $L=8$ down to 1.1×10^{-3} at $L=11$. The initial condition is a random-phase superposition of all modes with $|\mathbf{k}|\le2$, normalised to zero mean and unit variance.

A separable velocity such as $\mathbf{u}=(\sin y,\sin x)$ would be simpler and equally incompressible, but it cannot serve as a test. The advective and conservative forms differ by θ times the *discrete* divergence $\mathsf{D}_x u_x+\mathsf{D}_y u_y$, and for any $\mathbf{u}=(h(y),\,g(x))$ each of those terms vanishes separately: D_x applies an x-stencil to a field with no x-dependence and returns exactly zero. The two forms then reduce to the same expression, measured as $\|\mathrm{adv}-\mathrm{cons}\|_\infty/\|\mathrm{adv}\|_\infty=6.0\times10^{-15}$. In the cellular flow the same two terms are each $O(1)$, reaching 1.000 in magnitude, and cancel only in sum to 1.9×10^{-14}; the transport forms then differ by 6.7×10^{-4}, an $O(\Delta x^2)$ difference that the skew form averages. Only the second kind of velocity can verify that the conservative form is implemented correctly or that the skew form conserves $\langle\theta^2\rangle$ better than the advective one.

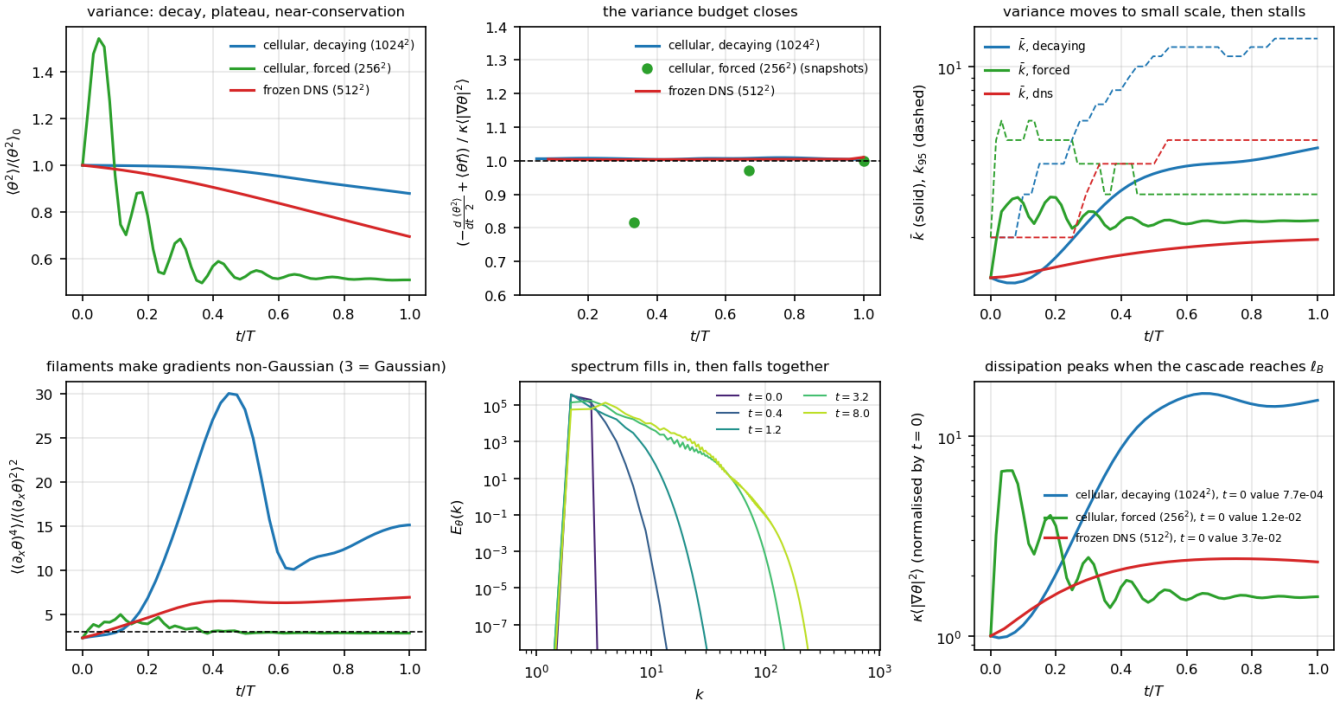


FIG. 1. Physical histories, from the dense reference. (a) Variance: monotone decay without forcing, a plateau with it. (b) The budget of Eq. (4) formed numerically; the unforced flows close it to within the line width, and the forced case approaches closure as the steady state is reached, evaluated at the stored snapshots since the production $\langle \theta f \rangle$ is not a recorded diagnostic. (c) Centroid $\bar{k}$ and the 95% wavenumber k_{95} move to small scale and stall, which is the cascade reaching ℓ_B. (d) Gradient flatness rises well above the Gaussian value 3 as filaments form. (e) The variance spectrum filling in. (f) Scalar dissipation, peaking as the cascade arrives at ℓ_B.

Space is discretised with second-order centred stencils for both the transport and the diffusion terms, and the transport term is written in skew form, $\frac{1}{2}$(advective + conservative), which inherits the antisymmetry of the continuous operator and conserves $\langle \theta^2 \rangle$ to round-off in the absence of truncation. Time integration uses SSPRK3 with a step satisfying both the advective CFL condition and the two-dimensional diffusive limit $\Delta t \leq \Delta x^2 / 4\kappa$.

Resolution is varied at fixed $\ell_B/\Delta x = 2.88$ by scaling $\kappa \propto \Delta x^2$, anchored at $\kappa = 5 \times 10^{-3}$ for $L = 8$, so that each resolution resolves the same range of scales. The comparison ladder integrates to $T = 8$, chosen from rank saturation rather than from a decay time: the intrinsic rank of the exact solution plateaus at $t \simeq 5$–6 at every resolution, and the truncation error saturates on the same timescale, so $T = 8$ leaves about 40% margin. Because $\kappa \propto \Delta x^2$, the variance remaining at $T = 8$ is not comparable across the ladder — 0.50, 0.88 and 0.95 of its initial value at $L = 8$, 10 and 11 — whereas the rank plateau is. The two other flows run to $T = 60$ and $T = 2$ for the reasons given in Secs. IX and X; Table II lists all of them.

Both encodings are advanced by identical code. The standard tensor train is the case $N = 0$ of the hybrid encoder, verified to reproduce `tt.vector_to_mps(vec,2,2L)` bond for bond. Compression policy, contraction order, stencils, velocity, time step and dense reference are shared, so the head size is the only variable between the two.

III. TIME HISTORIES

Figures 1 and 2 collect the quantities recorded at every sample of every run. The first is the physics, taken from the dense reference and shown for all three flows so that a statement made for one can be checked against the others. The second is the encoding, drawn at one resolution and head size with the bond cap swept, since that is the only control.

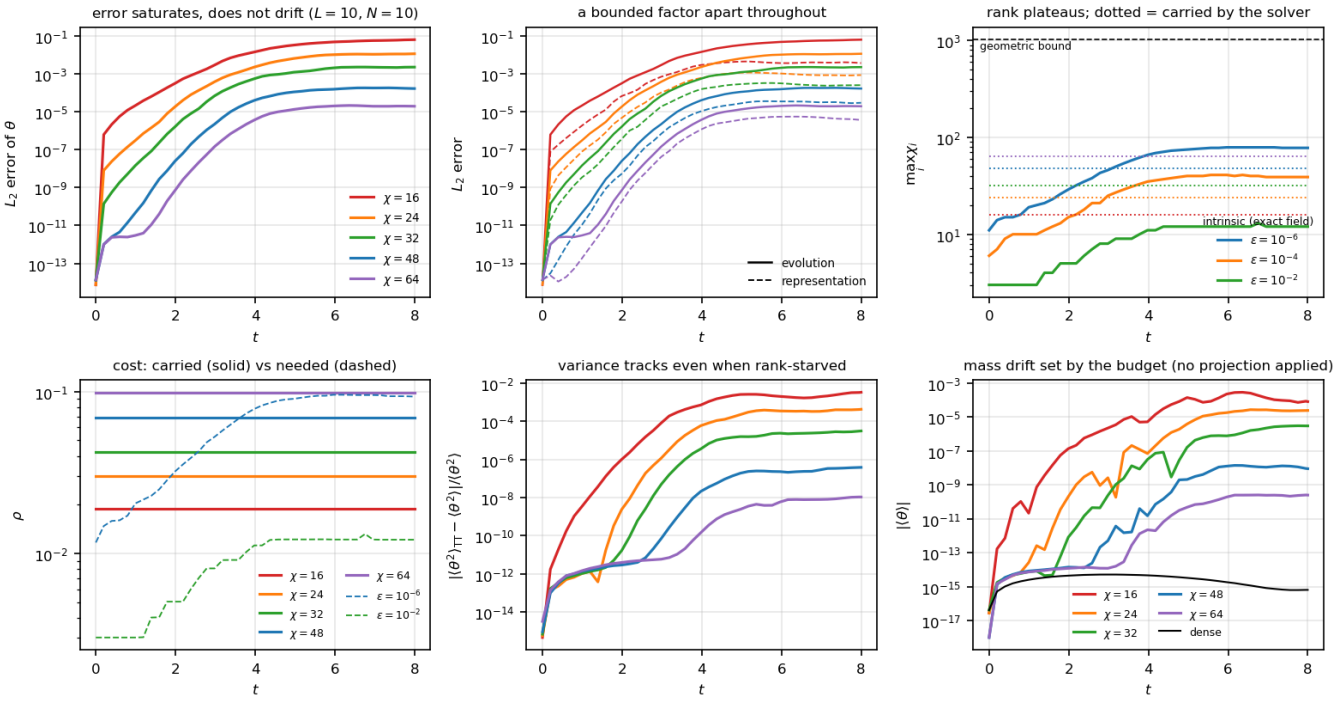


FIG. 2. Compression histories at $L = 10$, $N = 10$, bond cap swept. (a) The error rises while scales are being generated and then saturates rather than drifting. (b) The same curves against the error of merely representing the exact field at that budget, a bounded factor below. (c) Rank: the intrinsic rank of the exact solution at three probe accuracies (solid) plateaus far below the geometric bound, while the rank carried by the solver (dotted) sits at its cap throughout, ϵ being off. (d) Stored parameters, carried against needed. (e) The variance error, two to four orders of magnitude below the pointwise error at the same budget. (f) Mass drift: truncation is not mass-conserving and no projection is applied, so $|\langle\theta\rangle|$ grows by an amount set by the budget, against 10^{-15} for the dense reference.

IV. HEAD SIZE: WHAT N CONTROLS

The field is encoded on $2L$ binary legs in Morton order, even legs carrying the row index and odd legs the column index, with level 0 the most significant bit. The head fuses the first N legs into a single dense tensor of physical dimension 2^N; the remaining $2L - N$ legs are ordinary binary cores.

N therefore selects how many of the *most significant* bits of each axis are held densely. The head holds $N/2$ bits per axis, which is a $2^{N/2} \times 2^{N/2}$ grid of coarse blocks, each block spanning $2^{L-N/2}$ cells on a side; the structure inside a block is carried by the tail. Table III gives the correspondence.

N	bits/axis	blocks	cells per block		
	$N/2$	$2^{N/2} \times 2^{N/2}$	$L = 9$	$L = 10$	$L = 11$
4	2	4×4	128	256	512
6	3	8×8	64	128	256
8	4	16×16	32	64	128
10	5	32×32	16	32	64
12	6	64×64	8	16	32

TABLE III. Head size in terms of the grid it resolves densely. $N = 10$ at $L = 10$ holds 5 of the 10 bits of each axis, i.e. a 32×32 coarse grid, with the remaining 32×32 cells inside each block carried by the tail.

The useful range is $N \simeq L$, i.e. half the bits of each axis. The three sizes $N \in \{L-2,\ L,\ L+2\}$ bracket the optimum at every resolution measured, and the results below are reported on that bracket.

Two distinct normalisations of the head cost are worth separating, since they behave differently as L grows (Fig. 3). The head occupies $2^N \chi_0$ parameters. Relative to the compressed state this is a large and increasing share: at $L = 10$, $N = 10$, $\chi = 32$ it is 74% of the stored parameters. Relative to the dense field it is $2^N \chi_0 / 4^L$, which for $N \simeq L$ is

$\chi_0/2^L$ and therefore decreases with resolution: the same configuration stores a head worth 3.1% of the dense field, and the complete state worth 4.2%.

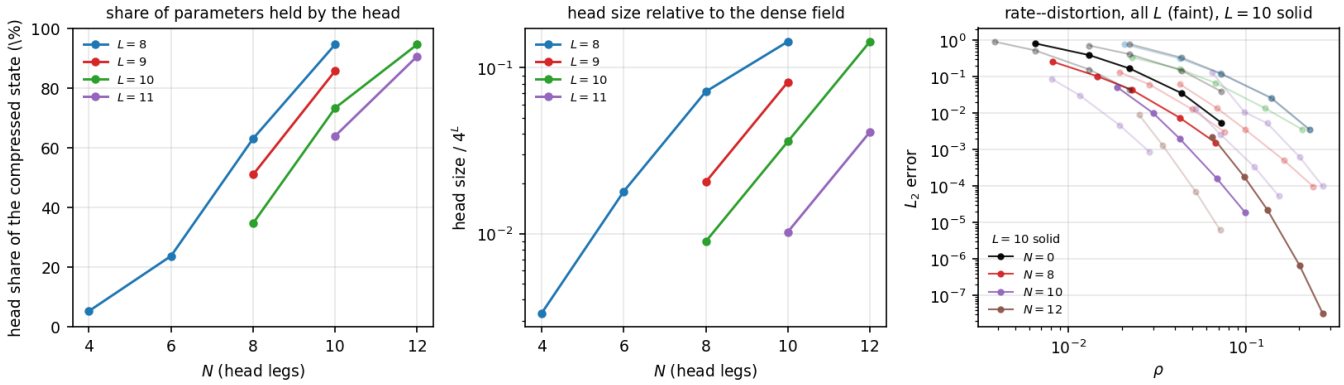


FIG. 3. Head cost under both normalisations, and the rate–distortion curves from which the optimum is read. Left: share of the compressed state held by the head. Centre: head size as a fraction of the dense field, which falls with L at fixed $N - L$. Right: L_2 error against compression ratio ρ for every run; $L = 10$ drawn solid.

V. RATE–DISTORTION

Bond caps are integers, so the two encodings do not land on identical parameter counts. Each encoding is therefore swept over bond caps and reported against the compression ratio $\rho = \text{parameters}/4^L$ it achieves. Table IV lists every run. The reported error is the relative L_2 difference from a dense reference integrated with the same scheme, step and velocity, averaged over the final third of the run, where it has saturated.

Alongside each evolution error is the error of compressing the *exact* field at the same budget, which separates the cost of representing the solution from the additional cost of evolving in the representation.

VI. PARAMETERS REQUIRED AT FIXED ACCURACY

The quantity of practical interest is the budget required to reach a given accuracy. Table V reports the ratio $\rho_{\text{std}}/\rho_{\text{hybrid}}$, with the head size that achieves the smaller budget, obtained by log–log interpolation of $\rho(\text{error})$. Targets are listed only where both encodings span the target error.

The error ratio read at fixed ρ instead of fixed accuracy is considerably larger — at $L = 10$ it runs from 2.8 at $\rho = 0.01$ to 25 at $\rho = 0.05$ — and it grows with resolution. Both quantities describe the same pair of curves. The rate–distortion curves are steep and steepen with L, so a horizontal offset of a factor 2 in ρ appears as one to two decades along the error axis. The budget ratio is the quantity that answers how many parameters the head buys.

VII. EVOLUTION AGAINST REPRESENTATION

Figure 4 collects the comparison across resolutions. The lower middle panel plots the evolution error against the representation error at the same budget for every run. All points lie between the 1× and 25× lines, with most below 5×, across four resolutions, five head sizes and five bond caps.

The practical consequence is that the ordering of encodings in N and the scaling in L can be established by compressing stored fields, which costs minutes, rather than by integrating them, which costs core-days. The representation measurement does not substitute for the evolution measurement when the absolute error level matters.

VIII. COST

Table VI gives measured cost per step at $\chi = 32$. At equal bond cap the hybrid encoding is slightly cheaper than the standard one, since it carries $2L - N$ tail cores instead of $2L$, while the head contributes $O(2^N r^2 \chi)$ through the

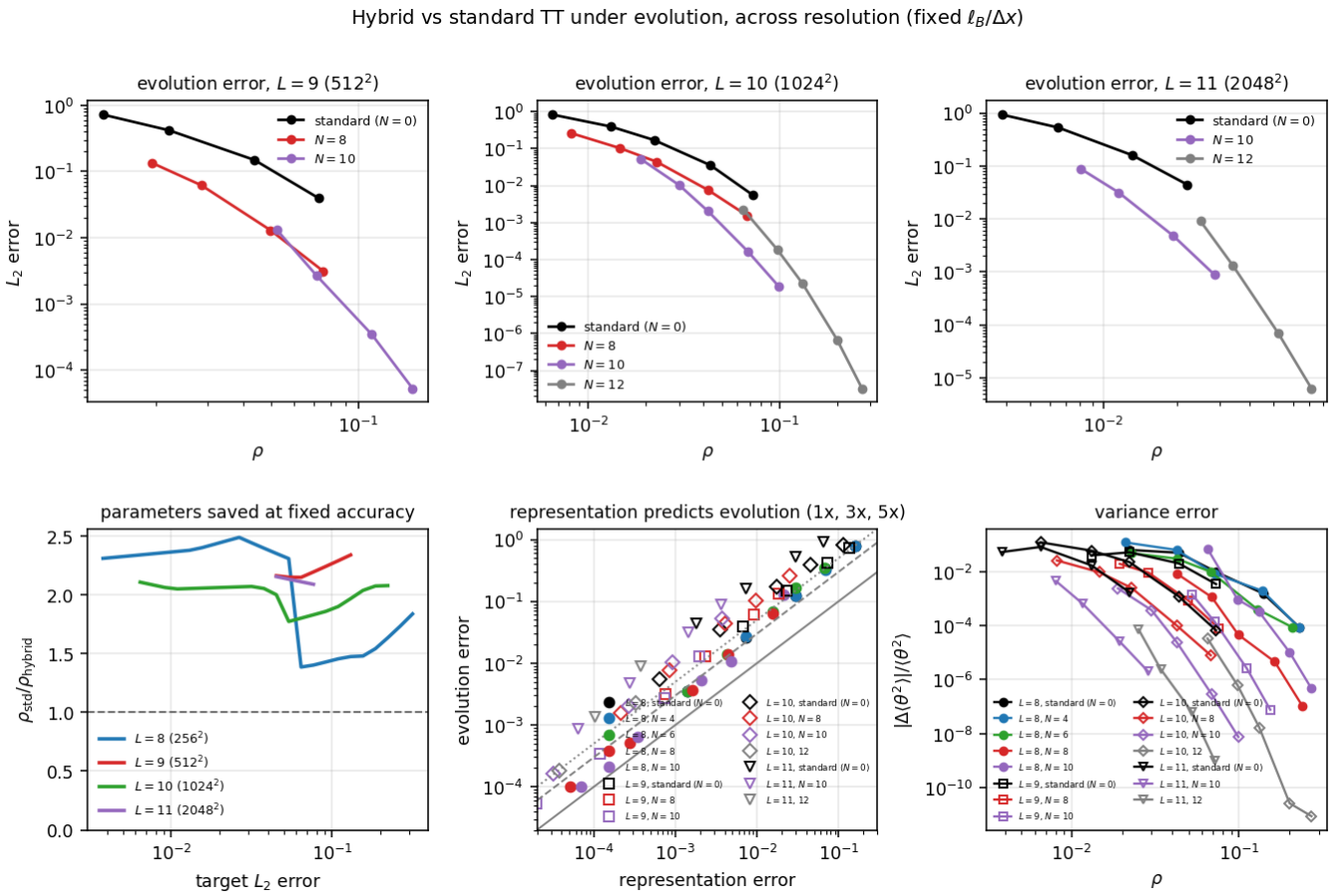


FIG. 4. Top: rate–distortion at $L = 9, 10, 11$. Bottom left: parameters saved at fixed accuracy. Bottom centre: evolution error against representation error at the same budget, with 1×, 3× and 5× guides. Bottom right: relative error in $\langle\theta^2\rangle$.

un-fusing path. At equal ρ the comparison is closer, because reaching a given ρ with a larger head permits a smaller χ.

All runs are single-threaded. On a 256-core node, four threads instead of one measured 29× slower per step, because a step consists of thousands of small contractions and SVDs, none large enough to amortise thread dispatch.

IX. FORCED STEADY STATE

The results above are obtained in a decaying cascade. To check that they are not an artefact of the transient, the cellular flow was rerun at 256^2 with a steady source $f = 0.8 \sin 4y$ to $T = 60$.

The forced wavenumber matters. The approach to a statistically steady state is governed by the diffusive time of the forced scale, $1/(\kappa k_f^2)$, which is 200 time units at $k_f = 1$; amplitudes from 0.02 to 0.05 were all still climbing at $t = 90$. At $k_f = 4$ that time is 12, and the variance plateaus by $t \simeq 40$, settling at 0.10, 0.22 and 0.51 for amplitudes 0.3, 0.5 and 0.8.

The forcing has zero box mean, so $\langle\theta\rangle$ is conserved: the dense reference holds it to 3×10^{-14} over 12 222 steps. Truncation is not renormalised and no mass projection is applied, so the hybrid solver instead drifts by an amount set by the budget, from 1.0×10^{-2} at $N = 0$, $\chi = 16$ to 1×10^{-13} at $N = 10$, $\chi = 32$. The variance is not conserved and is not expected to be: it rises to 1.51 at $t \simeq 4$ and settles at 0.5109, where production balances dissipation to 1.4% ($\langle\theta f\rangle = 1.895 \times 10^{-2}$ against $\kappa\langle|\nabla\theta|^2\rangle = 1.922 \times 10^{-2}$ at $t = 60$). What the hybrid encoding reproduces is that steady level, to 3×10^{-9} relative at $N = 8$, $\chi = 32$.

In the steady state the variance holds to within 1.3% beyond $t = 36$, and the intrinsic rank of the exact solution, probed at $\epsilon = 10^{-6}$ in the $N = 8$ layout, rises from 12 to a peak of 56 and settles at 41.7 ± 0.5 against a geometric bound of 256. The evolution error is monotone in χ for every head size, and the budget saving over the standard

$L = 8$ (256^2)

N	χ	ρ	head %	L_2 (evolution)	L_2 (representation)
0	8	0.0221	–	7.96×10^{-1}	1.62×10^{-1}
0	12	0.0436	–	3.35×10^{-1}	6.92×10^{-2}
0	16	0.0729	–	1.22×10^{-1}	2.93×10^{-2}
0	24	0.1393	–	2.64×10^{-2}	7.30×10^{-3}
0	32	0.2291	–	3.57×10^{-3}	1.41×10^{-3}
4	8	0.0208	9	7.79×10^{-1}	1.62×10^{-1}
4	12	0.0423	7	3.22×10^{-1}	6.92×10^{-2}
4	16	0.0716	5	1.24×10^{-1}	2.93×10^{-2}
4	24	0.1380	3	2.66×10^{-2}	7.30×10^{-3}
4	32	0.2278	2	3.57×10^{-3}	1.41×10^{-3}
6	8	0.0228	34	3.44×10^{-1}	6.79×10^{-2}
6	12	0.0423	28	1.65×10^{-1}	2.98×10^{-2}
6	16	0.0677	23	6.76×10^{-2}	1.54×10^{-2}
6	24	0.1282	18	1.38×10^{-2}	4.34×10^{-3}
6	32	0.2083	15	3.58×10^{-3}	1.41×10^{-3}
8	8	0.0423	74	6.40×10^{-2}	1.57×10^{-2}
8	12	0.0687	68	1.39×10^{-2}	4.38×10^{-3}
8	16	0.0989	63	3.63×10^{-3}	1.59×10^{-3}
8	24	0.1634	57	5.21×10^{-4}	2.73×10^{-4}
8	32	0.2396	52	9.85×10^{-5}	5.14×10^{-5}
10	4	0.0648	97	1.30×10^{-1}	2.15×10^{-2}
10	6	0.0981	96	1.09×10^{-2}	4.78×10^{-3}
10	8	0.1321	95	5.32×10^{-3}	2.08×10^{-3}
10	12	0.2005	94	6.41×10^{-4}	3.46×10^{-4}
10	16	0.2708	92	1.00×10^{-4}	6.87×10^{-5}

$L = 9$ (512^2)

N	χ	ρ	head %	L_2 (evolution)	L_2 (representation)
0	12	0.0131	–	7.23×10^{-1}	1.36×10^{-1}
0	16	0.0221	–	4.20×10^{-1}	7.39×10^{-2}
0	24	0.0436	–	1.50×10^{-1}	2.33×10^{-2}
0	32	0.0729	–	3.95×10^{-2}	6.65×10^{-3}
8	12	0.0194	61	1.35×10^{-1}	1.85×10^{-2}
8	16	0.0286	55	6.20×10^{-2}	9.17×10^{-3}
8	24	0.0496	47	1.30×10^{-2}	2.30×10^{-3}
8	32	0.0755	41	3.13×10^{-3}	7.48×10^{-4}
10	12	0.0523	90	1.30×10^{-2}	1.93×10^{-3}
10	16	0.0716	87	2.69×10^{-3}	7.28×10^{-4}
10	24	0.1112	84	3.47×10^{-4}	1.19×10^{-4}
10	32	0.1536	81	5.32×10^{-5}	1.98×10^{-5}

$L = 10$ (1024^2)

N	χ	ρ	head %	L_2 (evolution)	L_2 (representation)
0	16	0.0065	–	8.28×10^{-1}	1.15×10^{-1}
0	24	0.0131	–	3.96×10^{-1}	4.51×10^{-2}
0	32	0.0221	–	1.70×10^{-1}	1.73×10^{-2}
0	48	0.0436	–	3.54×10^{-2}	3.55×10^{-3}
0	64	0.0729	–	5.46×10^{-3}	6.41×10^{-4}
8	16	0.0081	48	2.62×10^{-1}	2.54×10^{-2}
8	24	0.0146	40	1.03×10^{-1}	9.67×10^{-3}
8	32	0.0228	34	4.41×10^{-2}	4.10×10^{-3}
8	48	0.0423	28	7.60×10^{-3}	8.50×10^{-4}
8	64	0.0677	23	1.56×10^{-3}	2.16×10^{-4}
10	16	0.0189	83	5.19×10^{-2}	3.77×10^{-3}
10	24	0.0300	78	1.02×10^{-2}	9.06×10^{-4}
10	32	0.0423	74	2.03×10^{-3}	2.66×10^{-4}
10	48	0.0687	68	1.62×10^{-4}	3.19×10^{-5}
10	64	0.0990	63	1.92×10^{-5}	4.68×10^{-6}
12	16	0.0648	96	2.27×10^{-3}	3.20×10^{-4}
12	24	0.0981	96	1.80×10^{-4}	3.71×10^{-5}
12	32	0.1322	95	2.26×10^{-5}	6.34×10^{-6}
12	48	0.2005	94	6.80×10^{-7}	2.46×10^{-7}
12	64	0.2708	92	3.26×10^{-8}	1.06×10^{-8}

$L = 11$ (2048^2)

N	χ	ρ	head %	L_2 (evolution)	L_2 (representation)

target L_2	$L=8$	$L=9$	$L=10$	$L=11$
1×10^{-1}	1.4 (N=6)	2.3 (N=8)	1.9 (N=8)	–
6×10^{-2}	2.3 (N=8)	2.1 (N=8)	1.8 (N=8)	2.1 (N=10)
3×10^{-2}	2.5 (N=8)	–	2.1 (N=10)	–
1×10^{-2}	2.4 (N=8)	–	2.1 (N=10)	–
3×10^{-3}	–	–	–	–

TABLE V. Parameters saved at fixed accuracy, and the head size achieving it. The saving is 1.8–2.5 and shows no trend with resolution over 256^2–2048^2.

L	N	χ	ρ	ms/step
8	0	32	0.2291	846
8	4	32	0.2278	920
8	6	32	0.2083	822
8	8	32	0.2396	535
9	0	32	0.0729	1507
9	8	32	0.0755	1419
9	10	32	0.1536	1518
10	0	32	0.0221	818
10	8	32	0.0228	940
10	10	32	0.0423	1068
10	12	32	0.1322	1989
11	0	32	0.0065	2257
11	10	32	0.0116	3348
11	12	32	0.0340	3596

TABLE VI. Measured cost per step at $\chi = 32$. Runs shared cluster nodes, so absolute values carry scheduling noise; the comparison within a row of constant L is the meaningful one.

encoding is 1.4 at $L_2 = 10^{-2}$, achieved at $N = 6$ (Table VII).

N	χ	ρ	head %	L_2 (evolution)	L_2 (representation)
0	16	0.0729	–	2.68×10^{-2}	3.14×10^{-3}
0	24	0.1393	–	2.86×10^{-3}	6.46×10^{-4}
0	32	0.2291	–	2.50×10^{-4}	1.05×10^{-4}
6	16	0.0677	23	1.22×10^{-2}	1.88×10^{-3}
6	24	0.1282	18	1.17×10^{-3}	4.28×10^{-4}
6	32	0.2083	15	2.50×10^{-4}	1.05×10^{-4}
8	16	0.0989	63	4.66×10^{-4}	2.21×10^{-4}
8	24	0.1634	57	5.68×10^{-5}	3.74×10^{-5}
8	32	0.2396	52	9.07×10^{-6}	6.35×10^{-6}
10	16	0.2708	92	1.32×10^{-5}	1.15×10^{-5}
10	24	0.4095	92	6.34×10^{-7}	5.20×10^{-7}
10	32	0.5521	91	5.14×10^{-8}	4.54×10^{-8}

TABLE VII. Forced cellular flow, 256^2, $f = 0.8\sin 4y$, averaged over the final third of the run.

X. A FROZEN DNS VELOCITY

The cellular flow is integrable and its scalar spectrum steep, so the benchmark was repeated with a velocity taken from the DNS dataset. The streamfunction was recovered from the stored v_x, v_y at 4096^2 by inverting the vorticity spectrally, coarsened to 512^2, and differentiated with the solver's own MPOs, leaving the velocity divergence-free to 2×10^{-12} discretely.

The streamfunction itself is close to full rank. At $\epsilon = 10^{-6}$ its bond saturates the geometric bound and the tensor train costs more parameters than the dense field, $\rho = 1.33$ at $L = 9$ and 2.32 at $L = 10$. It becomes compressible only at $\epsilon = 10^{-2}$, giving bond 21 at $\rho = 0.084$ for a 1.7% error in ψ. The runs therefore use a truncated velocity, carrying

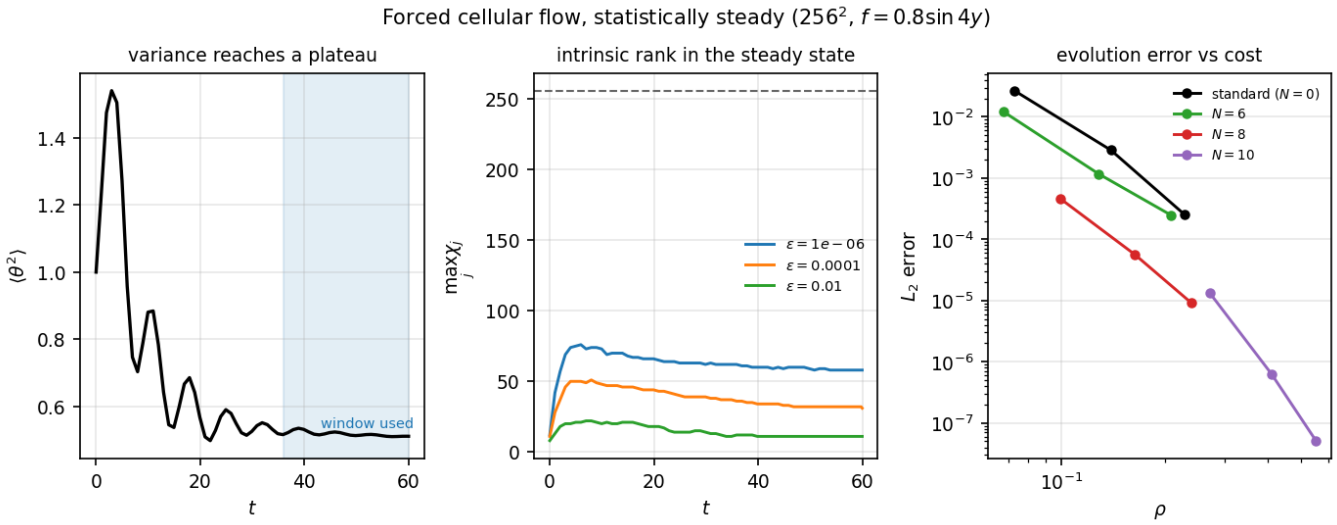


FIG. 5. Variance plateau, intrinsic rank in the steady state, and rate–distortion for the forced cellular flow.

$\chi(\mathbf{u}) = 34$ against 4 for the cellular flow, at $\rho(\mathbf{u}) = 0.14$, which makes a step about five times more expensive. The measured strain rate is $\gamma = 14.6$ and $u_{\max} = 1.78$, so $\kappa = 1.5 \times 10^{-2}$ places the Batchelor scale at $\ell_B/\Delta x = 2.61$, close to the 2.88 of the cellular ladder, and $T = 2$ covers 29 strain times.

The rank plateaus here as well, and faster: it is reached within a tenth of a time unit and then holds at 315 ± 1.1 at $\epsilon = 10^{-6}$ (61% of the geometric bound), 154 at 10^{-4} and 34 at 10^{-2}.

The evolution error, however, ceases to be monotone in the bond cap. At $N = 10$ the L_2 error runs 7.7×10^{-2}, 1.5×10^{-1}, 2.9×10^{-2}, 1.9×10^{-3} for $\chi = 16, 24, 32, 48$, and the variance error is non-monotone at $N = 8$, 10 and 12. The representation error at the same budgets is monotone in every case (Table VIII), which locates the effect in the dynamics rather than in the encoding. The ratio of evolution to representation error widens from 1.5–25 for the cellular flow to 1.5–233 here.

Two consequences follow. The rate–distortion framing used above does not transfer directly, since a given (N, χ) cannot be ranked by its budget alone; and the budget saving can be quoted at only one accuracy target where the two encodings overlap, 2.1 at $L_2 = 10^{-1}$, achieved at $N = 8$.

N	χ	ρ	head %	L_2 (evolution)	L_2 (representation)
0	16	0.0221	–	3.06×10^{-1}	5.94×10^{-2}
0	24	0.0436	–	1.82×10^{-1}	2.97×10^{-2}
0	32	0.0729	–	1.25×10^{-1}	1.37×10^{-2}
0	48	0.1393	–	6.61×10^{-2}	5.03×10^{-3}
8	16	0.0286	55	1.51×10^{-1}	1.29×10^{-2}
8	24	0.0496	47	8.60×10^{-2}	6.46×10^{-3}
8	32	0.0755	41	8.44×10^{-2}	3.83×10^{-3}
8	48	0.1341	35	3.40×10^{-2}	1.85×10^{-3}
10	16	0.0716	87	7.00×10^{-2}	2.62×10^{-3}
10	24	0.1112	84	1.34×10^{-1}	1.26×10^{-3}
10	32	0.1536	81	2.68×10^{-2}	7.22×10^{-4}
10	48	0.2396	78	1.84×10^{-3}	3.26×10^{-4}
12	16	0.2552	98	9.06×10^{-2}	3.89×10^{-4}
12	24	0.3836	98	4.47×10^{-4}	1.54×10^{-4}
12	32	0.5130	97	1.14×10^{-4}	5.13×10^{-5}
12	48	0.7669	98	7.41×10^{-6}	5.04×10^{-6}

TABLE VIII. Frozen DNS velocity, 512^2, ψ truncated at 10^{-2}, averaged over the final third of the run.

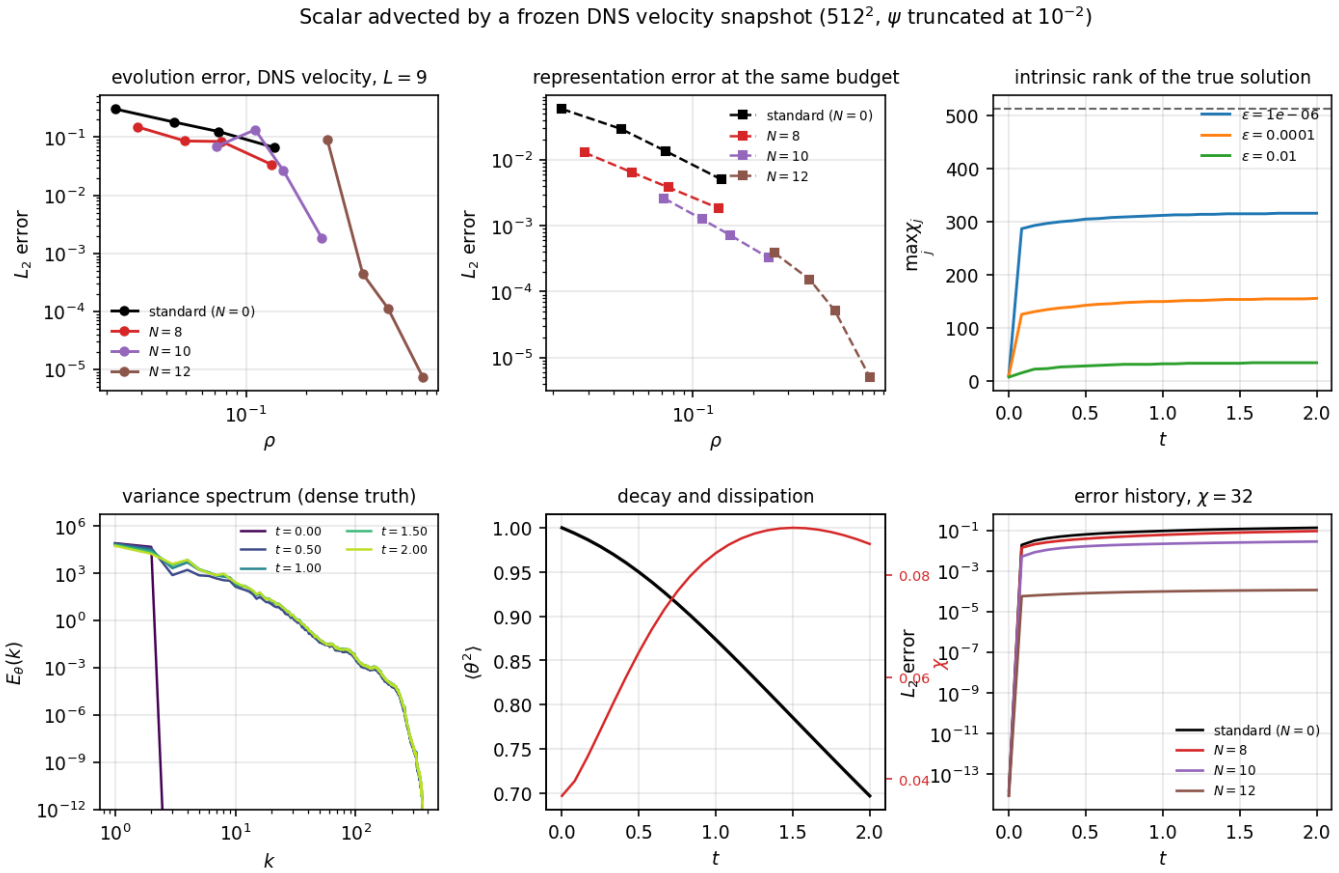


FIG. 6. Scalar advected by a frozen DNS velocity snapshot: rate–distortion, representation error at the same budgets, intrinsic rank, variance spectrum, variance budget and error histories.

XI. SCOPE OF THESE RESULTS

The measurements above are specific to the flow used. The cellular flow $\psi = \sin x \sin y$ has closed streamlines and is integrable, and its scalar spectrum is steep, $E_\theta(k) \sim k^{-4.9}$ over $3 \le k \le 20$, rather than the k^{-1} Batchelor form of a chaotic flow. Breaking the streamlines with $\psi = \sin(x + B \sin \omega t) \sin y$ raises the rank plateau from 75 to 84 at $L = 8$ and leaves the spectrum steep.

The size of the hybrid advantage depends on the field. On random-phase fields of prescribed spectral slope the same measurement gives 1.2 at $k^{-5/3}$ and 3.5 at k^{-5}, against 5–24 at fixed ρ for the cellular field. The difference is attributable to spatial coherence rather than to spectral slope or to alignment with the dyadic block boundaries: rigid shifts of the field by 2, 4, 8 and 13 cells leave the advantage unchanged.

Real turbulent scalar fields lie between the two cases. The DNS-velocity run above gives the first direct measurement on a realistic field: the advantage persists, at 2.1 in budget at $L_2 = 10^{-1}$, but the non-monotonicity of the evolution error means a single number characterises it less completely than in the cellular case. The representation measurements on DNS data reported in the paper remain the reference point for the representation question.

One further check bears on those figures. `tt.svd` defaults to `renormalize=True`, which rescales the kept singular values at every split. On `theta.00059` at 4096^2 the stored hybrid fields were found to equal the plain rank-truncated field times a single global scalar c to 1.2–3.5×10^{-14}. The norm is preserved exactly, $E(1)$ shifts by exactly c^2 — $+2.7\%$ at $\rho = 0.014$, falling to $+0.04\%$ at $\rho = 0.64$ — and the gradient flatness ratio is 1.0000, so all scale-invariant statistics are unaffected.

XII. CONFIGURATION

- Grids $2^L \times 2^L$ with $L = 8, 9, 10, 11$; $\kappa = 5 \times 10^{-3}$, 1.25×10^{-3}, 3.125×10^{-4}, 7.8125×10^{-5}.

- $\Delta t = 0.2\,\Delta x$ (CFL-limited at every resolution), $T = 8$, giving 1630, 3259, 6519 and 13038 steps.
- Head sizes $N \in \{0, 4, 6, 8, 10, 12\}$, bond caps $\chi \in \{4, \ldots, 64\}$ applied uniformly as `bd_first` = `bd_remaining`, with `eps` = `None` (so the cap alone truncates θ) and `renormalize` = `False` so that truncated norm is not restored.
- 101 production runs: 69 cellular decaying (29 at $L = 8$, 12 at $L = 9$, 20 at $L = 10$, 12 at $L = 11$), 12 cellular forced and 16 under the frozen DNS velocity. The $L = 10$, $L = 11$, forced and DNS sets were run as Slurm arrays, one core per task.